

Research Assignment 1

Section A : Data fundamentals

1. - Database $\Rightarrow$ an organised collection of data, which is usually stored in a computer system electronically.

$\rightarrow$ The data can be easily accessed, stored and organised for future use.

(Oracle, 2020)

- Databases are essential in the modern world because organisations generate and rely on large amounts of information every day. They improve efficiency, reduce duplication of information and allowing organisations to securely manage information across different systems.

Real world examples of database include :

a) Credit facilities : They use database to store and manage customer information such as contact details, loan advance amount, interest charged as well as fees.

b) Restaurants : They use database to store information on food stocks, expenses and revenue as well as customer preferences.

2. Data Warehouse $\Rightarrow$ It is a controlled system that collects and stores large amounts of data from different sources within an organisation.
(Oracle, 2023)

Regular Database

- Generally used for transactional purposes such as updating customer information or process payments

Data Warehouse

- It uses extract, transform and load (ETL) process to clean, prepare and organize for data analytics use. (IBM, 2025)

- A company uses a data warehouse to combine data from different systems in one place, making it easier to analyse trends, generate reports and support decision making. (Oracle, 2023)

e.g. Credit providers can track customer historical payments to identify repayment patterns.

3. - Data Lake is a centralised repository that allows you to store data (structured and unstructured) at any scale (AWS, 2026).

- The main difference is that a data warehouse stores processed and structured data, while a data lake stores large amounts of raw data.

- A business may use data lake when it needs to store large and various amount of data for future analysis

4. - Data Mart is a smaller, focused collection of data designed for a specific department or group of users. (IBM, 2025)

- A datamart is a focused slice of the datawarehouse

S Type of data	Explanation	Example
Structured data	Data that is highly organised and follows a fixed format, usually arranged in rows and columns (Oracle, 2022)	Employee ID
Semi-structured data	Does not allow follow a strict table but contains some organised content	Customer record
Unstructured data	Does not have a predefined structure.	PDF document

6. - Data Modeling is the process of creating a structured representation of the data that a system needs to store, manage and analyse.

e.g. Relational and dimensional Data Model.

7. - Data mining is the process of examining large amounts of data to identify patterns, relationships, trends and useful information (IBM, 2024).

8. A database management system (DBMS) is software used to store, organise, manage and update data in a database

DBMS	Description	Suitable use case
MySQL	An open source relational DBMS that uses SQL & stores data in tables (Oracle, 2026)	E-commerce web.
PostgreSQL	An open source object-relational DBMS that supports SQL	Financial systems

Oracle Database	A commercial relational DBMS that supports SQL.	Bank use it to manage financial transactions.
Mongo DB	A NoSQL, DBMS that stores data in a flexible document	Notion web.

9. ETL stands for Extract, Transform, Load.

- Used to collect data from different sources, clean and prepare it, and load it into a central data warehouse

10. Data storage formats are ways of organizing and storing data

- CSV - (Comma Separated Values) $\Rightarrow$ a simple text based format that stores data in rows and columns.
- JSON (JavaScript Object Notation) $\Rightarrow$ a text based format that stores data as key value pairs and can represent nested & hierarchical structures.
- Parquet (Apache Parquet) $\Rightarrow$ Columnar storage format
- Avro (Apache Avro) - row based data serialization format.

11. - Artificial Intelligence is the ability of computer systems to perform tasks that normally require human intelligence (Russel, 2010)

Narrow AI (Weak AI)	General AI (Strong AI)
- AI system designed to perform a specific task or limited range of tasks	- It has the ability to perform wide range of intellectual tasks across areas. (Cave, 2019)
e.g. Volvo assistant	e.g. A hypothetical human level intelligent machine

12. - Machine Learning - a branch of AI that enables computers to learn from data and improve their performance on a task without being explicitly programmed with every rule needed to perform that task. (Mitchell, 2021).

- In traditional programming, a programmer writes explicit instructions and rules that the computer follows to produce an output.

13. - Deep Learning - specialised area of machine learning that uses artificial neural networks.

14. - Natural language processing is a branch of AI that enables computers to understand and generate human language. (Jurafsky, 2023),
e.g. voice assistant, chatbots and customer service and autocorrect and predictive text.

15. - Large Language Model (LLM) is a type of AI model that is trained on large amounts of text data to understand and generate human language.

- The well known LLMs are Generative Pre-trained Transformer and Gemini.

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